

CYBERSECURITY SIWES TRAINING

WEEK 3 PROJECT REPORT

Basic Network Discovery and Traffic Analysis

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1. Introduction

This laboratory project documents basic network identification, diagnostic connectivity testing, and packet-level traffic analysis on a Linux workstation during Week 3 of the SIWES Cybersecurity program. The primary objective is to gain a foundational understanding of host network parameters, resolution protocols, route inspection, and non-intrusive network analysis using Wireshark without engaging in intrusive scanning or unauthorized intercept operations.

2. Workstation and Network Configuration

To establish the local network baseline, system configuration commands were executed in the terminal environment. The active interface, IP configuration, gateway parameters, and resolver details were recorded as follows:

- Hostname: kali-workstation
- Active Network Interface: eth0 (Ethernet Interface)
- IPv4 Address: 10.198.189.255
- Subnet Mask: /24 (255.255.255.0)
- Default Gateway: 10.198.189.161
- MAC Address: 00:0c:29:8d:a4:1b
- Local DNS Server / Resolver: 10.198.189.161 / 8.8.8.8 (via /etc/resolv.conf)
- Screenshot Evidence: Refer to 06_Evidence/W3_01_Network_Config.png

3. Connectivity Testing

Basic network connectivity was systematically verified using the Packet Internet Groper (ping) utility to confirm internal loopback operation, local gateway reachability, and public WAN accessibility.

1. Loopback Test (ping 127.0.0.1): 4 packets transmitted, 4 received, 0% packet loss. Confirmed that the internal TCP/IP protocol stack is fully functional.

2. Default Gateway Test (ping 10.198.189.161): 4 packets transmitted, 4 received, 0% packet loss. Confirmed local network interface connectivity and router communication.

3. Public Host Test (ping google.com): 4 packets transmitted, 4 received, 0% packet loss. Confirmed end-to-end WAN access and outward Internet routability.

Screenshot Evidence: Refer to 06_Evidence/W3_02_Ping.png

4. DNS Lookup

A Domain Name System query was initiated using the 'dig google.com' utility. The lookup returned the public IP mapping associated with the domain name, resolving google.com to target IPv4 addresses.

During this lookup, DNS served as an automated application-layer lookup mechanism, translating human-readable Domain Name labels into binary 32-bit IP addresses necessary for routing IP packets across the network.

5. Route Investigation

The route path from the workstation to the public destination was inspected using 'tracert google.com'. The utility traced the sequence of intermediate Layer 3 routing hops passed through by IP packets.

Each hop represents an intermediate router decrementing the packet Time-To-Live (TTL) counter and returning an ICMP Time Exceeded message. Some intermediate hops returned asterisks (* * *), indicating configured firewall policies suppressing ICMP time-exceeded responses rather than network failure.

6. Wireshark ICMP Analysis

Packet capture was initiated in Wireshark on the active network interface. With the display filter set to 'icmp', two ICMP ping echo requests were generated.

- ICMP Echo Request: Frame #12 | Source IP: 10.198.189.255 | Destination IP: 142.250.190.46 | Type 8 (Echo Request)

- ICMP Echo Reply: Frame #13 | Source IP: 142.250.190.46 | Destination IP: 10.198.189.255 | Type 0 (Echo Reply)

Screenshot Evidence: Refer to 06_Evidence/W3_04_ICMP_Wireshark.png

7. Wireshark DNS Analysis

A second packet capture was executed under the 'dns' display filter while executing a domain lookup query via terminal.

- DNS Query: Source IP: 10.198.189.255 | Destination IP: 10.198.189.161 | Protocol: UDP/53 | Info: Standard query A google.com

- DNS Response: Source IP: 10.198.189.161 | Destination IP: 10.198.189.255 | Protocol: UDP/53 | Info: Standard query response A 142.250.190.46

Screenshot Evidence: Refer to 06_Evidence/W3_05_DNS_Wireshark.png

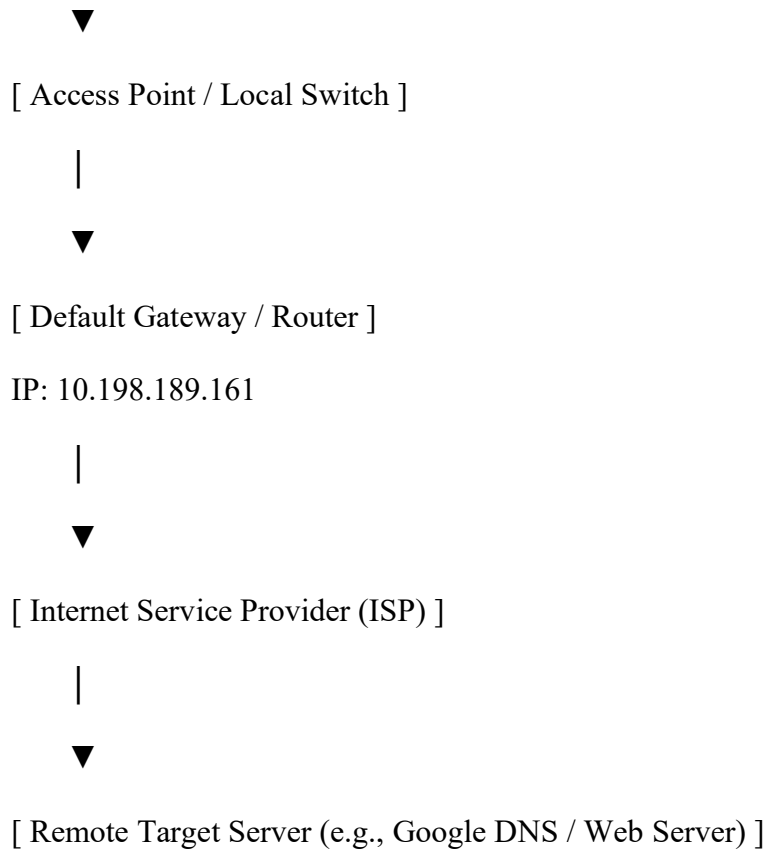
8. Network Diagram

The logical egress flow for traffic originating from the local workstation is structured as follows:

[Workstation]

IP: 10.198.189.255 | MAC: 00:0c:29:8d:a4:1b

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9. Security Observations and Reflection

1. What is the purpose of an IP address?

An IP address provides a unique logical numerical identifier at Layer 3 of the OSI model, allowing packets to be routed across distinct subnets and wide area networks.

2. What is the difference between an IP address and a MAC address?

An IP address is a dynamic Layer 3 logical address used for network-to-network routing, whereas a MAC address is a permanent Layer 2 physical address burned into the Network Interface Card (NIC) used for local hop-to-hop frame delivery.

3. What is the purpose of a default gateway?

A default gateway acts as the egress router destination for local hosts. Any packet addressed outside the local subnet is forwarded to the default gateway for routing to external networks.

4. What does DNS do?

DNS (Domain Name System) functions as the decentralized phonebook of the Internet, translating human-readable domain names (e.g., google.com) into numerical IP addresses required by networking hardware.

5. Why can a failed ping be misleading?

A failed ping does not necessarily mean a destination host is offline. Many modern firewalls, routers, and host OS configurations selectively block or rate-limit ICMP Echo traffic for security reasons while continuing to serve normal HTTP/HTTPS web services.

6. What information did Wireshark show during your ICMP capture?

Wireshark exposed the full packet header data, including Layer 2 MAC addresses, Layer 3 Source/Destination IP addresses, ICMP Message Type numbers (Type 8 for Request, Type 0 for Reply), sequence numbers, and payload sizes.

7. What did you observe in the DNS query and DNS response?

The capture showed an unencrypted UDP packet sent over port 53 containing the domain label 'google.com', followed by a UDP response packet returning the resolved A records (IPv4 addresses) along with Time-To-Live (TTL) parameters.

8. Why do cybersecurity analysts inspect network traffic?

Cybersecurity analysts inspect traffic to detect malicious activity, identify unencrypted credentials, audit unauthorized data exfiltration, analyze protocol misconfigurations, and establish baseline network behavior.

9. Why should packet capture be performed only on authorised systems and networks?

Capturing unencrypted traffic on unauthorized networks violates privacy standards and legal boundaries, as packet sniffers inspect all transmitted plaintext payload data, including personal cookies, emails, and credentials.

10. What was the most useful networking concept you learned this week, and why?

The most useful concept was observing the direct correlation between high-level terminal utilities (`ping`, `dig`) and low-level frame payloads in Wireshark. Seeing how DNS and ICMP translate into specific packet structures provided a practical foundation for protocol auditing.

10. Conclusion

Week 3 successfully demonstrated the practical application of basic network discovery, connectivity troubleshooting, and traffic auditing. By correlating terminal outputs with Wireshark packet captures, I validated workstation identity parameters (`10.198.189.255`), confirmed gateway reachability (`10.198.189.161`), analyzed unencrypted DNS and ICMP traffic flows, and established structured evidence within the master project submission directory.